

BEAVER ISLAND, MI, USA

WMO: 722186

Lat: 45.689N

Lon: 85.566W

Elev: 666

StdP: 14.35

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04892

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.1	3.8	-8.3	3.6	0.5	-3.5	4.7	5.3	25.5	26.6	23.7	26.0	4.4	40	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	82.0	68.7	79.6	67.2	77.4	66.1	71.4	78.6	70.0	76.2	68.5	74.6	9.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.2	110.1	75.4	67.9	104.9	73.9	66.2	98.8	72.4	35.6	78.9	34.4	76.4	33.1	74.6	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	19.4	17.5	DB	-9.4	87.4	8.3	1.9	-15.4	88.8	-20.2	89.9	-24.9	91.0	-31.0	92.4	(4)
(5)			WB	-10.0	73.9	8.1	1.5	-15.8	75.0	-20.5	75.8	-25.0	76.7	-30.9	77.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	22.5	21.4	29.3	40.2	51.5	60.7	66.3	66.2	60.0	48.7	38.0	28.5	(6)
(7)		DBStd	17.84	9.76	9.71	10.27	7.65	7.81	6.35	5.56	5.34	7.04	7.95	7.62	7.99	(7)
(8)		HDD50	3846	853	800	647	307	74	3	0	0	5	124	367	666	(8)
(9)		HDD65	7704	1318	1220	1108	745	424	158	52	49	180	509	810	1131	(9)
(10)		CDD50	1866	0	0	4	12	121	325	507	502	306	83	6	0	(10)
(11)		CDD65	250	0	0	0	0	6	30	93	86	31	4	0	0	(11)
(12)		CDH74	1471	0	0	1	0	73	212	603	439	135	8	0	0	(12)
(13)		CDH80	192	0	0	0	0	8	32	98	42	12	0	0	0	(13)
(14)	Wind	WSAvg	7.9	9.0	8.5	7.8	8.2	7.5	6.5	6.4	6.4	7.0	8.5	9.1	9.5	(14)
(15)	Precipitation	PrecAvg	33.2	2.7	1.7	2.0	2.7	3.0	2.7	2.8	2.8	3.9	3.7	2.8	2.8	(15)
(16)		PrecMax	39.2	5.3	2.8	4.8	6.2	5.2	6.6	5.4	5.4	8.6	6.7	5.3	6.3	(16)
(17)		PrecMin	27.8	0.6	0.3	0.3	0.8	1.1	0.9	1.1	0.6	1.0	1.5	0.9	0.4	(17)
(18)		PrecStd	3.2	1.3	0.8	1.4	1.5	1.1	1.5	1.2	1.2	1.8	1.6	1.3	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	45.6	66.6	69.7	81.1	84.1	86.6	81.7	73.5	63.0	52.3	(19)	
(20)			MCWB	42.1	42.1	57.3	54.3	65.5	69.1	69.7	70.5	68.9	64.1	54.8	50.0	(20)
(21)		2%	DB	40.9	39.5	53.9	62.8	75.0	79.3	82.3	81.0	77.9	69.9	56.5	44.6	(21)
(22)			MCWB	38.3	36.6	47.0	50.3	61.8	66.3	68.9	69.0	67.1	62.6	51.7	42.0	(22)
(23)		5%	DB	37.3	36.3	46.4	57.3	70.3	76.1	79.9	78.8	74.9	65.3	52.3	41.7	(23)
(24)			MCWB	35.3	34.1	41.3	47.2	58.5	63.8	67.3	67.2	66.0	59.2	48.0	39.3	(24)
(25)		10%	DB	35.4	34.1	42.6	53.8	65.8	72.8	77.3	76.7	72.3	61.5	49.7	39.0	(25)
(26)			MCWB	34.0	32.2	37.4	44.5	55.4	61.9	66.2	66.2	64.6	56.2	45.7	36.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.5	42.8	58.4	57.0	67.5	71.9	73.6	73.1	71.5	67.8	55.9	50.2	(27)
(28)			MDCB	45.2	45.3	66.1	65.4	77.2	80.8	81.8	80.1	78.2	72.9	60.9	51.8	(28)
(29)		2%	WB	38.4	37.6	47.7	52.0	64.2	68.5	71.2	71.5	69.5	63.7	52.0	42.4	(29)
(30)			MDCB	40.6	40.0	51.9	60.2	72.3	76.2	78.7	78.2	74.4	67.3	55.9	44.1	(30)
(31)		5%	WB	35.7	34.5	41.4	48.3	60.5	66.2	69.5	70.0	67.9	59.8	48.6	39.6	(31)
(32)			MDCB	37.3	36.0	46.6	56.1	68.2	73.6	76.3	75.4	72.9	64.7	51.9	41.7	(32)
(33)		10%	WB	33.7	32.4	37.8	45.3	57.1	63.9	68.2	68.3	65.9	56.6	46.1	37.1	(33)
(34)			MDCB	35.0	33.7	41.7	52.0	63.7	70.7	74.9	73.9	71.1	61.5	48.7	38.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	11.9	14.3	16.3	19.8	19.3	18.5	17.2	17.0	13.1	10.4	8.8	(35)
(36)			MCDBR	10.4	12.0	19.0	24.3	25.8	22.5	20.9	18.8	18.0	17.2	15.1	11.6	(36)
(37)			MCWBR	9.8	10.7	13.3	14.6	14.2	11.9	10.4	9.9	9.6	11.5	11.7	11.0	(37)
(38)		5% WB	MCDBR	9.9	11.6	18.7	22.0	22.9	19.2	17.2	15.5	15.4	15.8	14.4	11.6	(38)
(39)			MCWBR	9.5	10.9	13.7	14.9	14.5	11.7	9.9	9.2	10.5	11.8	12.0	11.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.285	0.290	0.315	0.352	0.381	0.392	0.389	0.383	0.368	0.339	0.305	0.280	(40)	
(41)		taud	2.458	2.404	2.399	2.360	2.361	2.369	2.405	2.419	2.444	2.495	2.525	2.511	(41)	
(42)		Ebn at Noon	264	284	291	287	281	278	277	275	269	263	254	255	(42)	
(43)		Edh at Noon	22	28	32	36	38	38	36	34	31	25	20	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	364	659	1107	1459	1791	1941	1955	1659	1259	734	389	271	(44)	
(45)		RadStd	40	68	115	152	150	140	79	104	102	85	54	26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page